FPGA Tactical Controller for StarCraft II Micro Scenarios

EECS 3216 — Updated proposal draft, November 12, 2026

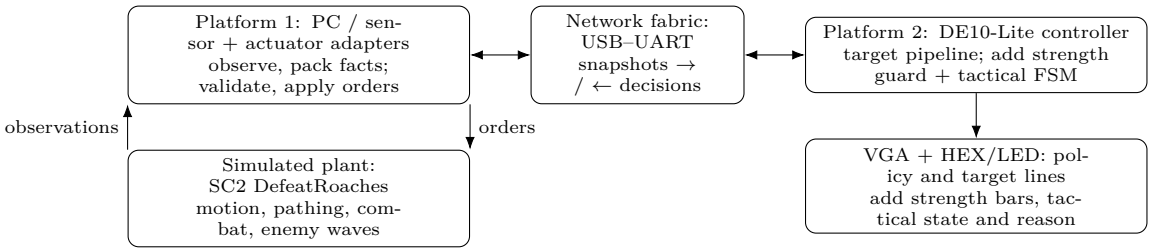
Target: Tier A, subject to instructor approval; existing demo provisionally approved by TA

Group: [Lead name/ID], [Development partner name/ID], [Report partner name/ID], [Demo partner name/ID]

1. Starting point and bounded objective

We will extend the working DE10-Lite demonstrator in PR #6: live SC2 *DefeatRoaches*, Marines versus Roach waves, at most 16 units/side. The PC sends observations over UART; the FPGA chooses every Marine's target. `SW0` selects CLOSEST or FOCUS (weakest in range, otherwise nearest); VGA shows markers and target lines. The lead reports a defeat-to-victory demo; retained results establish software mean scores of 32.1/62.4 over ten matched seeds and board CLOSEST parity. Record board FOCUS comparisons before claiming a hardware outcome improvement. The project lead implemented the work to date.

2. Feedback architecture: a simulated cyber-physical plant



Adapted from Lee–Seshia Fig. 1.1 [1]. Sensors/actuators are API adapters; SC2 substitutes for a physical plant. The PC transports facts and translates FPGA decisions; SC2 executes movement, firing and damage. Built-in auto-targeting can occur between decisions.

3. Highest-value algorithms and observable FSM behavior

- **Required foundation: nearest versus weakest-in-range focus fire [2].** Preserve integer ranking, squared distances and deterministic index ties. Show one saved snapshot under both switches, then reset matched live episodes. This controls *who* receives ATTACK orders; a policy toggle is not itself an ATTACK/RETREAT transition.
- **Planned extension: Lanchester-inspired strength guard [3].** For homogeneous forces per side, compute $P_F = k_F N_F^2$ and $P_E = k_E N_E^2$, where $k_F \propto \mathrm{DPS}_F/\mathrm{HPmax}_E$ and conversely for k_E. Use fixed integer coefficients, a 17-entry square lookup, widened products/comparisons; no ODE solver, division or square root. Add friendly HP and explicit encounter reset. Proposed guard: retreat if $4P_F < 3P_E$ or friendly HP is at most 25% of encounter-initial maximum. Latch RETREAT until reset; HOLD within one map unit of an authored safe point. Otherwise ATTACK when enemies exist; HOLD when none. VGA adds strength bars, state and guard reason. Freeze bounds, coefficients, widths, transition priority and move/hold semantics before HDL. Partial health, range, geometry and waves limit the square-law model. Judge retreat by survival, separately from victory/score.
- **Defer:** damage reservations (no gain in the existing software sweep), target hysteresis and cooldown kiting. No mixed armies, learning, search or general navigation.

4. Hardware value, scope and verification

MAX 10 `10M50DAF484C7G`, 50 MHz; custom UART RX/TX (115200, 8-N-1), checksum/parser FSMs, register records, three-stage distance/ranking pipeline, reply FSM and 640×480 VGA using a 25 MHz pixel enable. No RAM/PLL IP is required by the existing design. PR #6 reports 1,216 reference-matching cases in each simulator and board CLOSEST tests, 32% LEs and +4.283 ns setup slack. Reported core maximum: 258 clocks (5.16 μs); round trip median: 15.9 ms. These are baseline results, not new measurements or CPU-speedup evidence. Demonstrate concurrent communication/control/display, predictable core cycles, switch-selected decisions and error recovery. Check both policies and new state/guard sequences against an independent reference; refit/timing-check extensions. Use step mode, eight game loops/decision; log versions, seeds, score/outcome, survivors, errors and separate core/link latency. Gate live retreat integration by Nov. 20; if blocked, seek assessed-scope revision and label the strength display advisory.

5. Realistic ownership and delivery

Lead: architecture, algorithms/FSM, protocol/SC2 integration, reviews and final sign-off. **Development partner:** own board, UART and VGA; independent setup, strength-bar/status display and reference/board regression assistance; lead owns integration. **Report partner:** references, architecture explanation, evidence tables and report assembly. **Demo partner:** scenario setup, matched-run evidence, cabling/checklist and timed rehearsal/narration. Names remain unassigned; these are future deliverables, not retrospective equal coding credit. Freeze by Nov. 26; evaluate/rehearse Nov. 27–Dec. 1; demo/report Dec. 2.

[1] E. A. Lee and S. A. Seshia, *Introduction to Embedded Systems*, 2nd ed., Fig. 1.1, p. 6; Sec. 3.3 (2017). [2] D. Churchill and M. Buro, *Portfolio Greedy Search and Simulation for Large-Scale Combat in StarCraft*, Sec. III (2013). [3] M. Stanescu, N. Barriga and M. Buro, *Using Lanchester Attrition Laws for Combat Prediction in StarCraft*, AIIDE (2015). Our guard is an engineering adaptation, not their learned model.